UNIVERSITY OF WATERLOO



MTE 544 – Lab 1 Conclusions

By:

Hynes, Aidan (21015093)

McPhee, Michael (21021347)

McIntyre, Will (21009235)

Using:

Station 5, Robot 17

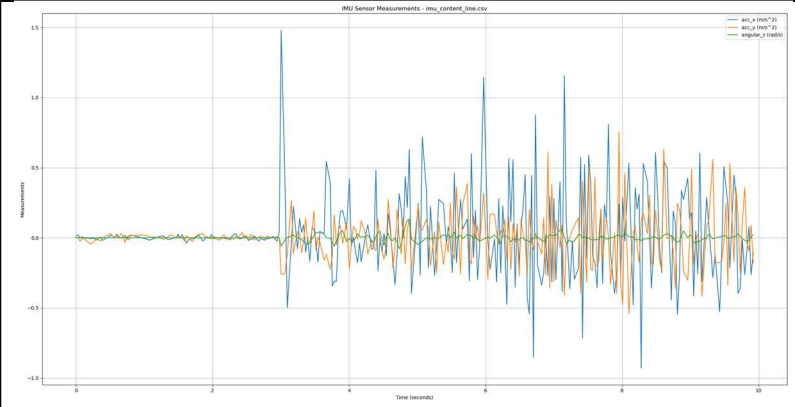


Figure 1: IMU Sensor Measurements for TurtleBot with Linear Trajectory

The angular velocity during linear acceleration stays around zero as expected. The linear acceleration profiles show an initial spike from the inertia of starting the robot to move. There are also sharp spikes where the control loop increases the speed, and high frequency noise which could be from the robot vibrating against the floor.

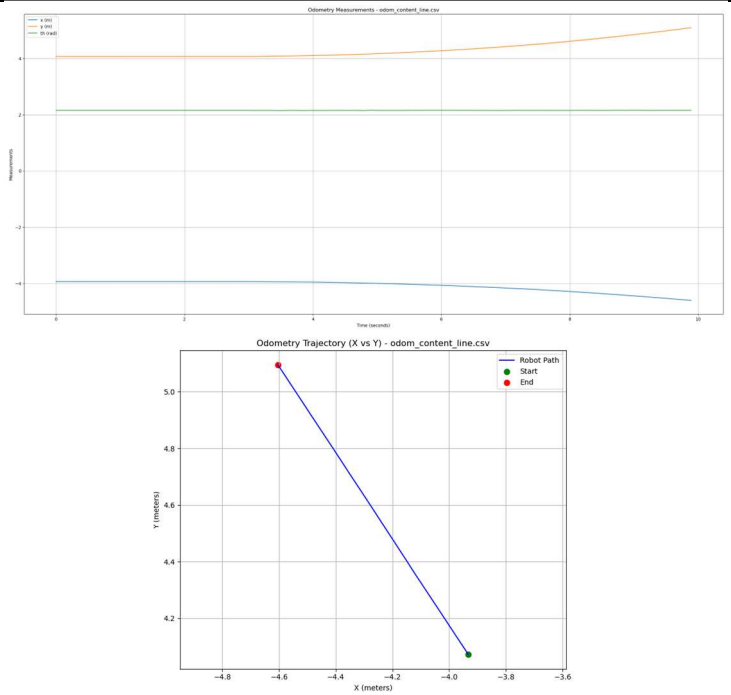


Figure 2: Odometry Measurements for TurtleBot with Linear Trajectory

The odometry plot for the straight-line, constant-acceleration test shows the pose of the robot over time. The robot starts near (x, y) = (-4, 4) and accelerates in the negative x direction and the positive y direction. Both coordinates change quadratically with time, as expected. The heading angle θ remains constant at roughly 2.2 rad.

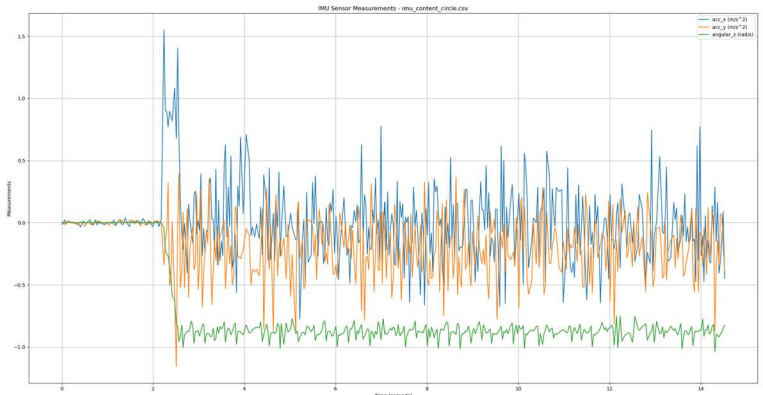


Figure 3: IMU Sensor Measurements for TurtleBot with Circular Trajectory

There is a spike from inertia to get the robot to move. Once moving, the angular velocity drops to around -0.9 rad/s reflecting constant rotational velocity around circular path. The linear acceleration components once again display noise, but the acceleration in the y component is consistently lower than that of the x, which is centered around 0 $\frac{m}{s^2}$. The centripetal acceleration v^2/r in circular motion is the acceleration in the y component of the robot's frame.

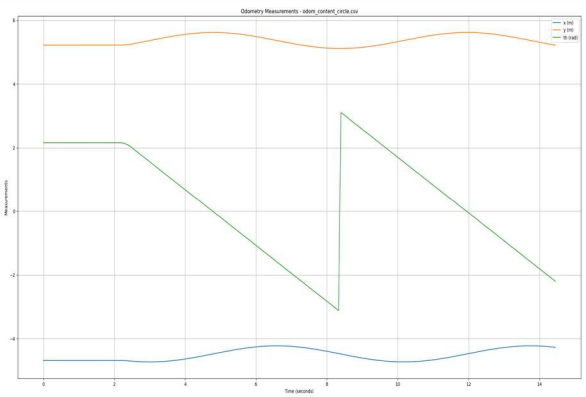
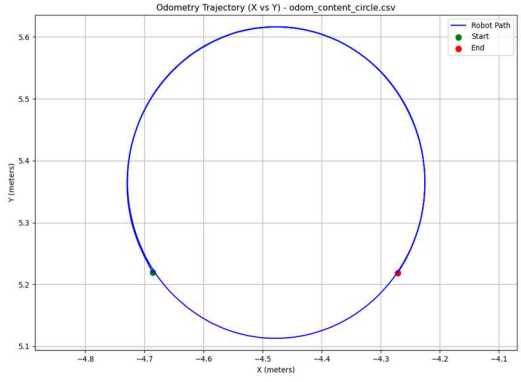


Figure 4: Odometry Measurements for TurtleBot with Circular Trajectory



When the robot moves in circular motion, the x and y positions exhibit smooth, sinusoidal oscillations from their starting positions. The orientation angle θ decreases linearly as the robot turns continuously in a loop and the vertical discontinuity at t = 8.2 seconds represents the angular wrapping where the angle transitions from $-\pi$ to $+\pi$ rad.

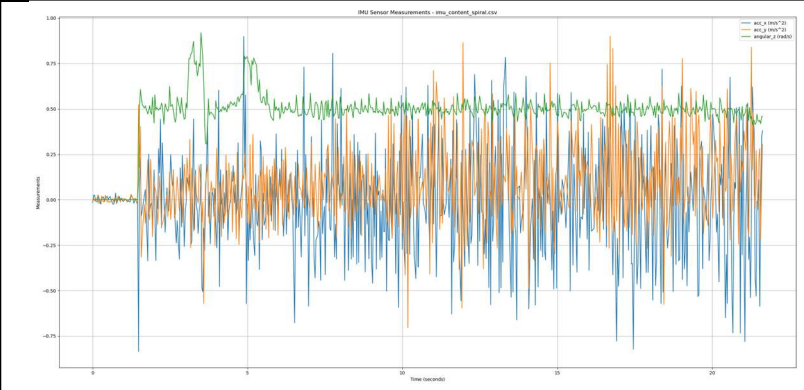


Figure 5: IMU Sensor Measurements for TurtleBot with Spiral Trajectory

There is once again an initial stationary phase followed by a spike in x acceleration. Once spiral motion has begun, the angular velocity stays relatively constant around 0.5 rad/s as expected based on our code. There is again a large amount of noise in the linear acceleration components caught by the IMU. The acceleration in the y component of the robot reference frame increases linearly from centripetal acceleration $v^2/r = v\omega$ (v increasing linearly, ω constant).

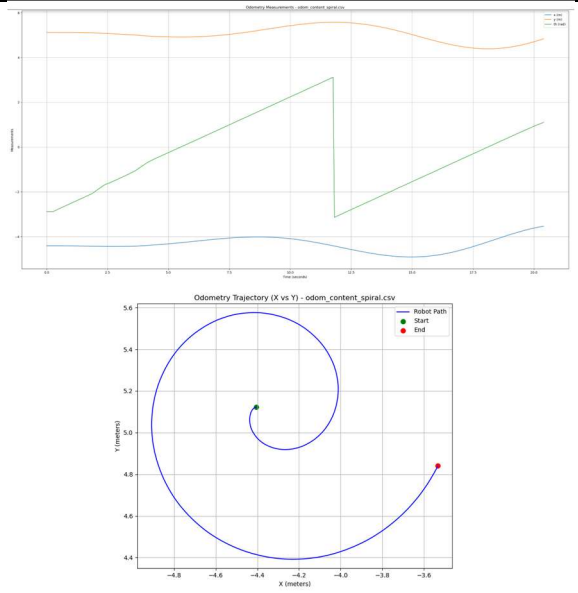


Figure 6: Odometry Measurements for TurtleBot with Spiral Trajectory

When the robot moves in spiral motion, the x and y positions exhibit smooth, sinusoidal oscillations with increasing amplitude. The orientation angle θ increases linearly as the robot turns continuously in a loop and the vertical discontinuity at t = 11 seconds represents the angular wrapping where the angle transitions from π to $-\pi$ rad.

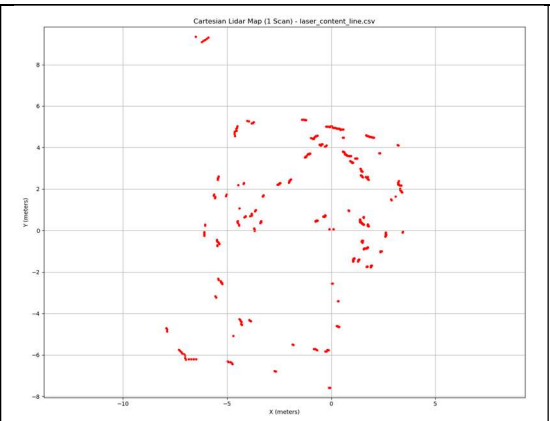


Figure 7: Cartesian Lidar Map for a Single Scan After Line Trajectory

The plot shows a single 2D laser scan projected in Cartesian coordinates (X and Y meters) after linear motion. It is difficult to make out exact features due to scattered points in the data as we were in a large room with many other students and robots. However, there is a general shape of the lab.

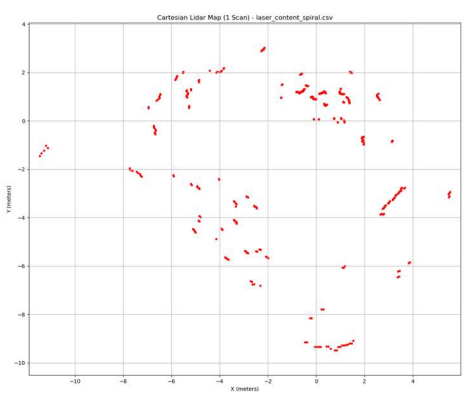


Figure 8: Cartesian Lidar Map for a Single Scan After Spiral Trajectory

The plot displays a single 2D laser scan this time after a spiral trajectory. There are some noticeable gaps in the data but the plot still shows some sweeping arc segments indicating wall segments.

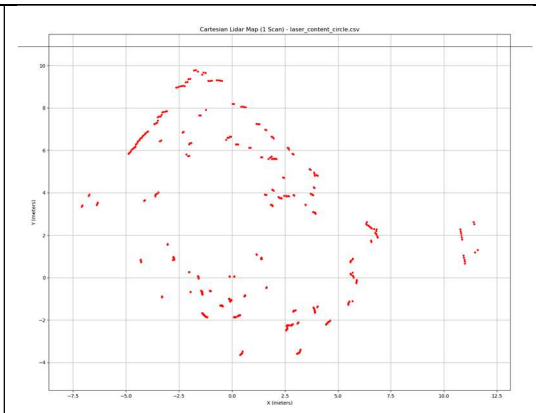


Figure 9: Cartesian Lidar Map for a Single Scan After Circular Trajectory

The plot once again shows a single laser scan captured after circular motion. This plot shows the clearest features with continuous series of points making up interpretable environment features.

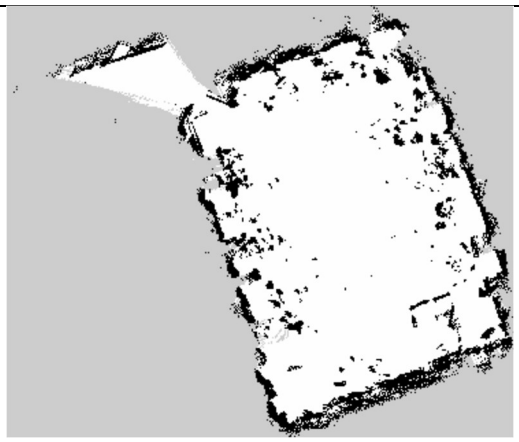


Figure 10: Nav2 SLAM Occupancy Grid Map of the Room

The plot displays the occupancy grid map generated by the robot with Nav2 while using teleop to drive around the parameter of the room. The white areas represent the open space, and the black regions indicate obstacles or room boundaries. The map captures the general shape and boundaries of the room reasonably well, particularly along the outer edges. The partially uneven and scattered markings suggest some noise in the sensor readings, which may be caused by sensor limitations, reflections, and moving objects or people within the room. Overall, the result shows that Nav2 was able to collect sensor data and build a recognizable representation of the room. However, the scattered points and rough boundaries suggest that the map could be improved through combining additional sensor data, smoother robot movement, and repeated passes around the perimeter to improve accuracy and consistency.